

Cloud Service Models: IaaS, PaaS, and SaaS

IaaS - Infrastructure as a Service

IaaS provides virtualized computing infrastructure over the internet. The cloud provider manages the physical hardware, while the user manages everything from the operating system upward — including runtime, middleware, applications, and data.

Key Features

- Virtual machines — rent virtual servers instead of buying physical hardware
- Storage on demand — block, object, and file storage available instantly
- Networking control — configure firewalls, load balancers, and IP addresses
- OS freedom — install any operating system (Windows, Linux, etc.)
- Pay-per-use — billed based on consumption (hourly/monthly)
- High customization — full control over environment configuration
- Global availability — data centers spread across multiple regions

Responsibility Model

You manage: OS, runtime, middleware, apps, data

Provider manages: servers, storage, networking, virtualization

Who Uses IaaS

- IT administrators and system engineers
- Startups needing scalable infrastructure without upfront cost
- Enterprises migrating from on-premise data centers

EXAMPLE : IaaS — Amazon EC2 (Elastic Compute Cloud)

Amazon EC2 provides virtual servers (instances) in the cloud. Companies like Netflix use it to run multiple VMs for large-scale applications. Users can choose the OS, install software, configure security, and scale instances based on demand. It follows a pay-as-you-use model and gives full control over the system.

PaaS - Platform as a Service

Provides a ready-to-use platform for developers to build, test, and deploy applications — without managing the underlying infrastructure.

Key Features

- Faster development — no infrastructure setup needed
- Built-in tools — databases, middleware, dev tools
- Auto-scaling managed by the provider
- Ideal for developers who want to focus on code

Responsibility Split

You manage: application code and data

Provider manages: OS, runtime, servers, storage

Who Uses PaaS:

- Software developers and application programmers building web or mobile apps
- DevOps engineers setting up CI/CD pipelines and automated deployments
- Startups wanting to launch products quickly without managing servers
- Data scientists and ML engineers using managed notebooks and model training platforms

EXAMPLE: PaaS — Google App Engine

Google App Engine allows developers to upload code while Google manages servers, scaling, and OS maintenance. For example, a startup can deploy an application and make it live quickly. The platform automatically handles traffic increases without manual effort, so developers focus only on coding.

SaaS - Software as a Service

Fully hosted software delivered over the internet. Users simply log in and use the app — no installation, no maintenance required.

Key Features

- Zero installation — accessible from any browser
- Automatic updates managed by the provider
- Subscription-based pricing (monthly/annual)
- Best for end users — no technical setup

Responsibility Split

You manage: your data and user access only

Provider manages: everything — app, OS, servers, storage

Who Uses SaaS:

- Business executives and managers using CRM tools like Salesforce
- Students and teachers using Google Workspace or Microsoft 365 for learning
- HR and finance teams using payroll and accounting software like QuickBooks
- Marketing teams using email and campaign tools like Mailchimp

EXAMPLE : SaaS — Zoom

Zoom provides ready-to-use video conferencing software over the internet. Companies simply subscribe and start using it without building infrastructure. Zoom manages servers, updates, and performance, while users access it through a browser or app.